

SHREYAA LAKSHMIVARAGAN

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Dear Hiring Team,

I am excited to apply for the AI Automation Engineering Intern position at DarkHaven AI. As a recent Master's graduate in Computer Science from Binghamton University, I have focused my academic and project work on building practical AI systems that combine machine learning, Large Language Models, backend engineering, and automation to solve real-world problems. The opportunity to build production-ready AI solutions alongside experienced engineers at DarkHaven AI is exactly the kind of environment I have been looking for.

Throughout my graduate studies, I have intentionally pursued projects that go beyond developing machine learning models by emphasizing software engineering, automation, and scalable system design. I enjoy breaking down complex workflows into reliable technical solutions, integrating AI into existing systems, and continuously improving processes through thoughtful automation.

One of my most significant projects is **DocMind**, an AI-powered document question-answering platform built using Python, OpenAI APIs, Retrieval-Augmented Generation (RAG), vector embeddings, and Streamlit. I designed a modular retrieval pipeline that transforms static documents into an intelligent knowledge base capable of semantic search and context-aware responses. Building this application required integrating AI APIs, designing effective prompts, managing structured data, improving retrieval quality, and continuously refining the system based on testing and performance. The experience strengthened my understanding of building reliable AI-powered applications rather than simply connecting to language models.

During my internship at Global Health Impact, I developed end-to-end Python data pipelines that automated the collection and processing of healthcare data across more than 200 countries using APIs. I designed workflows that standardized and validated data from multiple sources, significantly reducing manual effort while improving data quality for researchers. Working on these systems reinforced the importance of writing maintainable code, documenting processes clearly, and collaborating with stakeholders to translate operational needs into technical solutions.

I also developed a Social Media Sentiment Analytics platform using Python, PostgreSQL, SQL, Flask, REST APIs, and Postman that automated the ingestion and analysis of more than 245,000 social media posts. Through this project, I gained practical experience integrating APIs, working with relational databases, developing backend services, and building scalable data processing

workflows. I have consistently used Git for version control and enjoy following clean software engineering practices that make projects easier to maintain and extend.

Beyond my coursework, I enjoy exploring new AI technologies and experimenting with emerging tools. Whether testing new LLM capabilities, improving prompts, or evaluating different retrieval strategies, I enjoy understanding how AI systems can become more accurate, efficient, and reliable. I believe curiosity and continuous learning are essential in AI engineering, and I actively seek opportunities to expand my knowledge through hands-on experimentation.

What excites me most about DarkHaven AI is the opportunity to work on production systems where AI directly improves engineering workflows and operational efficiency. I am eager to contribute to designing AI-powered automations, integrating APIs, optimizing existing processes, and building solutions that create measurable value. I thrive in collaborative environments where ideas are shared openly, challenges are approached creatively, and learning is encouraged every day.

Thank you for considering my application. I would welcome the opportunity to contribute my experience with Python, AI APIs, machine learning, automation, backend development, and software engineering to your team. I look forward to discussing how my background, technical skills, and enthusiasm for AI automation can support DarkHaven AI's mission.

Sincerely,

Shreyaa Lakshmivaragan